



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Fumble

Group No : __, CSE370 Lab Section : __, Spring 2026		
ID	Name	Contribution
23201040	Mashrafi Haque	1. Startup creation (company) 2. Co-founder invitation 3. Job listing
23201490	Safwan Ahmed	1. chat/message 2. Social post 3. Investment
23201485	Imtiaz Al Shariar	1. log in/sign up 2. Match (swipe,search) 3. Profile 4. Job Application

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	4
3	ER/EER Diagram	5
4	Schema Diagram	5-6
5	Normalization	7
6	Frontend Development	8
7	Backend Development	9-10
8	Source Code Repository	11
9	Conclusion	11
10	References	

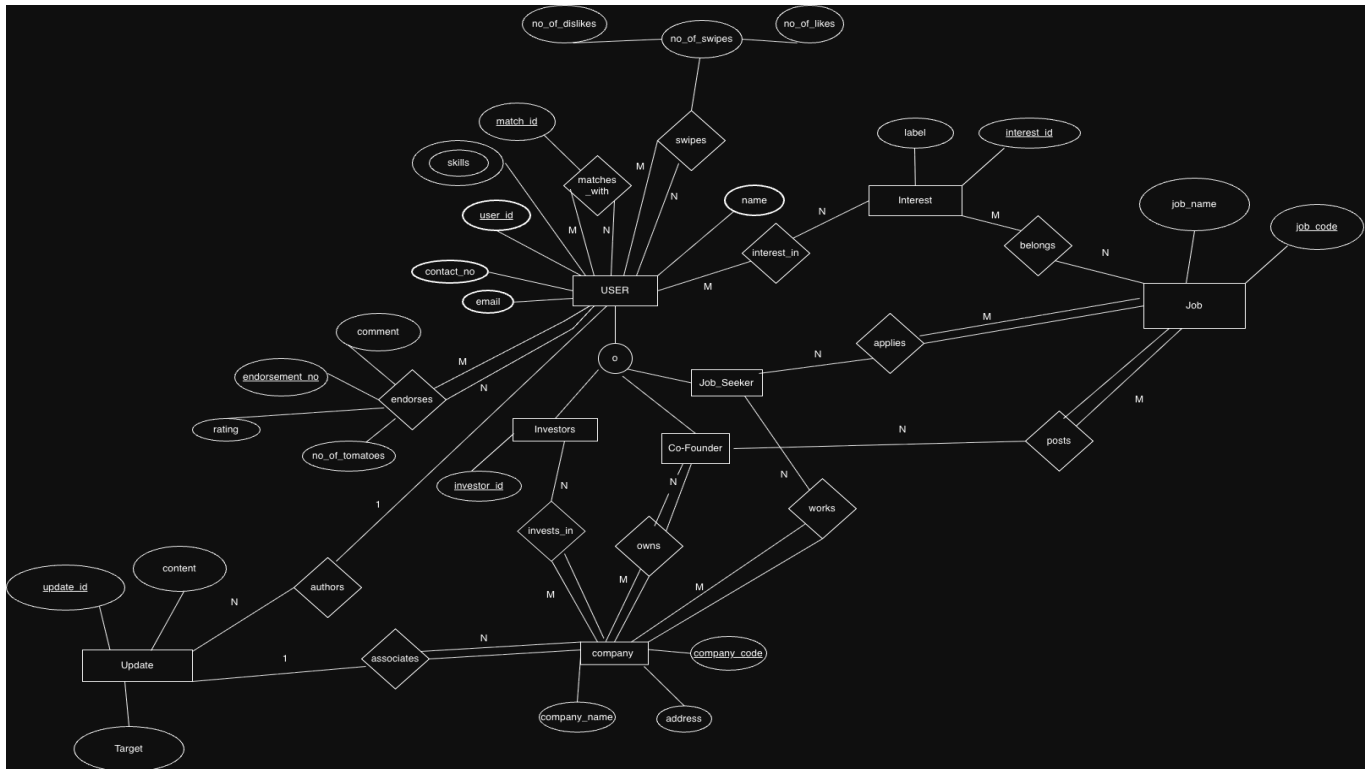
Introduction

Today a lot of people are willing to search for a job, founders are looking for co-founders to work on a project or run a company together. Moreover there are also people who want to invest in a company. The core idea of the project is to create a suitable platform for all of them. The platform that is developed came from the idea of modern dating apps like tinder, bumble. Basically if people are willing to work in a project they will simply right swipe and left swipe otherwise and afterwards a new job post will appear. Using this platform people can search for suitable collaborators or co-founder to work on a project or run a company. There will be options for investments in a company as well. The investors willing to invest can invest. Moreover, there are options for rating and reviewing user profiles and company profiles. This is the rough idea of our project.

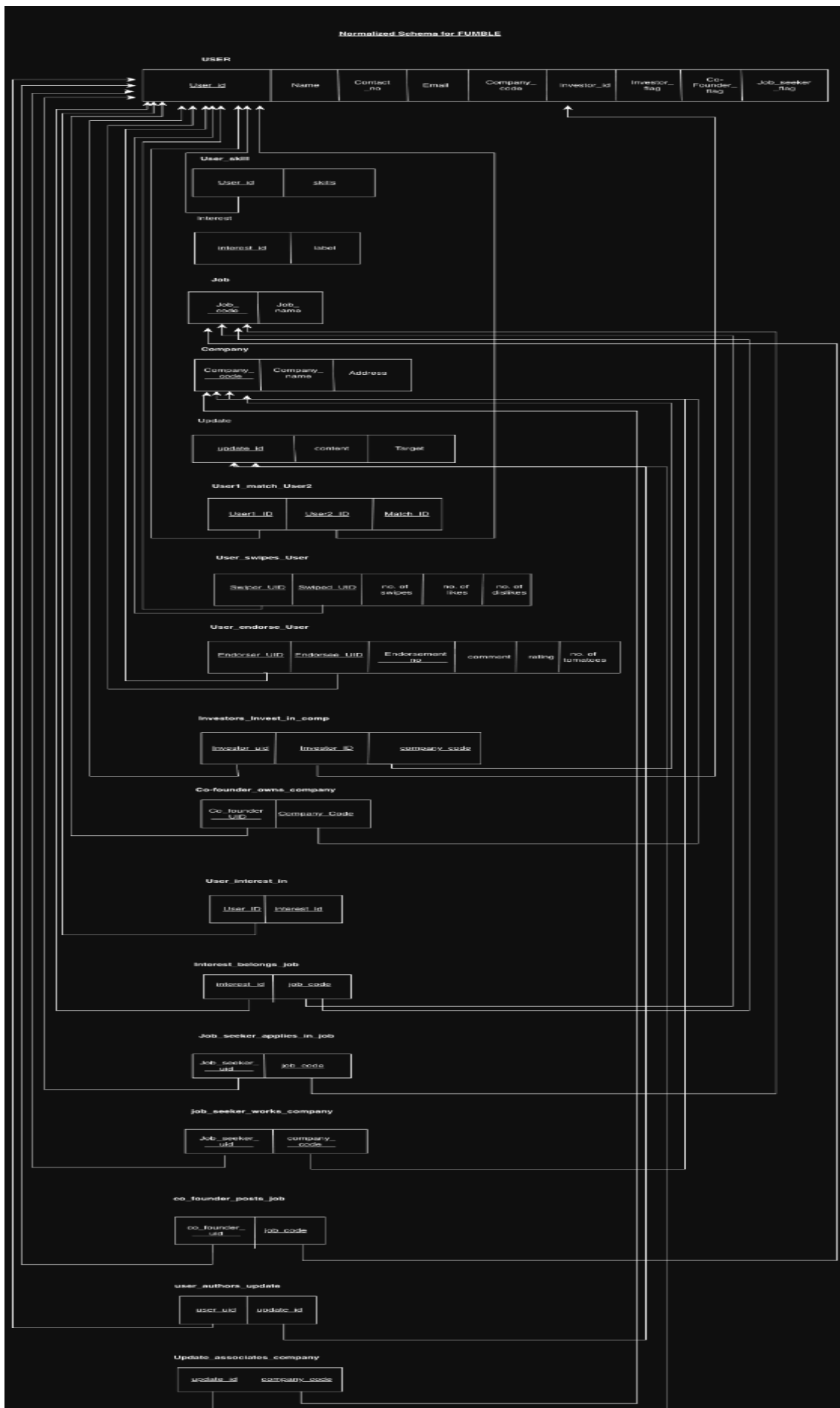
Project Features

ID, Name	Features [3 per member]	
23201040, Mashrafi Haque	Ft 1	Startup creation (company)
	Ft 2	Co-founder invitation
	Ft 3	Job listing
23201490, Safwan Ahmed	Ft 1	chat/message
	Ft 2	Social post
	Ft 3	Investment
23201485, Imtiaz Al Shahriar	Ft 1	log in/sign up
	Ft 2	Match (swipe,search)
	Ft 3	Profile
	Ft 4	Job Application

ER/EER Diagram



SCHEMA DIAGRAM



Normalization

The entire schema diagram is normalized to 3NF.

Frontend Development

Frontend Development Technology Stack

1. Framework: React 19 with TypeScript
2. Build tool: Vite (fast development and optimized builds)
3. Styling: Tailwind CSS v4 with a custom design system
4. UI components: shadcn/ui
5. Icons: Hugeicons (React integration)
6. Routing: React Router v7
7. State management: React Context API (Auth Provider)

Backend Development

TECHNOLOGY STACK

FastAPI — the main web framework that receives browser requests and sends back answers.

SQLAlchemy — helps our Python code talk to the database using simple objects instead of raw SQL everywhere.

SQLite — a small database file on the computer; good for a class project because it is easy to run.

python-jose and bcrypt — used for login security: passwords are hashed, and users get JWT tokens so the server knows who is logged in.

Pydantic — checks that incoming data looks correct before we use it (for example missing fields or wrong types).

Uvicorn — the program that runs the FastAPI server so the app stays online during testing.

HOW THE BACKEND IS ORGANIZED (LAYERS)

Think of four layers stacked from top to bottom:

Layer 1 — API (FastAPI): URLs, requests, responses, HTTP status codes.

Layer 2 — Business rules: signup/login, matching, important checks like “can this user do this?”

Layer 3 — Data access: SQLAlchemy models and database queries.

Layer 4 — Database (SQLite): where the rows and tables actually live.

MAIN DATABASE PIECES (IN PLAIN WORDS)

User — anyone who signs up (founders and job seekers). A user can have many posts and many messages.

Company — a startup profile. A company can link to multiple founders (users) and post multiple jobs.

Job — one job opening. Each job belongs to one company.

Swipe — a like or pass. It belongs to the person who swiped and points to a user or company they swiped on.

Message — a chat line. Each message has one sender and one receiver (both users).

CompanyInvitation — an invite to join a company as a co-founder. Links a company and two users (inviter and invitee).

Post — a feed post. Each post has one author (user).

KEY FEATURES (WHAT THE APP CAN DO)

1. LOGIN AND ACCOUNTS

How it works: passwords are stored safely (hashed). After login, the app gives a token so the user stays signed in for a while.

Main API paths: POST /auth/register — create an account POST /auth/login — log in and get a token GET /auth/me — see my profile PATCH /auth/me — update my profile

Basic safety ideas: passwords are hashed tokens expire after some time some routes require a logged-in user

2. SWIPING AND MATCHING

Main API paths: POST /swipes — save like or dislike GET /swipes/matches — see mutual likes GET /swipes/history — see past swipes

Simple flow: User A likes User B by swiping right. If User B already liked User A back, they become matched. Matched users can message each other.

3. MESSAGING

Main API paths: POST /messages — send a message GET /messages/{user_id} — view chat with one person GET /messages/conversations — list all chats PATCH /messages/{id}/read — mark a message as read

Simple rules: only matched users can message each other (this blocks random spam chats) tracks unread counts shows recent chats first messages can be marked read automatically when you open the chat

4. COMPANIES AND FOUNDERS

Main API paths: POST /companies — create a company GET /companies/my — list companies I belong to GET /companies/invitable — companies that still accept more founders (under the limit below) POST /companies/{id}/founders/{user_id} — add a founder DELETE /companies/{id}/founders/{user_id} — remove a founder

Important rules: a company cannot have more than five founders (enforced by the backend) only current founders can add or remove founders co-founder invites are tracked with CompanyInvitation records

5. JOBS

Main API paths: POST /companies/{id}/jobs — create a job under a company GET /jobs — list jobs GET /jobs/my — jobs linked to companies I belong to PATCH /jobs/{id} — update a job DELETE /jobs/{id} — delete a job

6. SOCIAL FEED (POSTS)

Main API paths: GET /posts — see the feed POST /posts — create a post GET /posts/{id} — get one post DELETE /posts/{id} — delete my own post

Source Code Repository

https://github.com/mashrafi19/CSE370_Project

Conclusion

Fumble is a small startup matchmaking app. It works a bit like a dating app: people swipe, get matches, and can chat. It helps founders find co-founders and job seekers find startups.

We built a website using typescript and a server using FastAPI (python) with a SQLite database. The app lets users sign in, make profiles, post updates, list companies and jobs, swipe, match, and message each other. Fumble is a learning project which shows how database ideas turn into a working app that fits a clear goal: connecting people around startups.